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Hybrid Multi-task Learning and Integrated Expert Labels For Advanced CNN-Based Document Forgery Detection

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Abstract— **The method of Document forgery has full-size demanding situations in daily life, impacting diverse sectors by way of compromising the authenticity by jeopardizing the legitimacy of important files. In a survey of over 400 lending professionals conducted by the fintech company Plaid, 61% of lending organizations identified themselves as victims of document fraud. Of these, 42% of cases involved fraudulent documents that had alterations in financial details such as bank balances, and transaction amounts, while 50% of documents had identity details altered (identity theft). This paper addresses this issue by offering a novel and a hybrid technique for forgery detection for both digital and handwritten documents using Convolutional Neural Networks (CNNs). Our method treats forgery detection as an image segmentation undertaking, leveraging the skills of CNNs to correctly pick out and delineate suspicious regions within any form of report, whether or not digital or handwritten. By posing forgery detection as segmentation trouble, we enhance the precision of our version in distinguishing between genuine and forged regions. In addition, to improve detection accuracy, our technique integrates multi-task mastering with the aid of incorporating auxiliary labels generated via a text detection model. This model produces bounding boxes around textual content, which can be used as extra labels at some point in training, permitting the CNN to study relevant functions for each text and forgery detection. The hybrid nature of our model guarantees robust overall performance throughout diverse report types, demonstrating its effectiveness in addressing the complexities of record forgery in sensible eventualities.**

Keywords— **Document forgery detection, Convolutional Neural Networks (CNNs), Image segmentation, Digital and handwritten documents, Multi-task learning, Text detection, Hybrid model.**

I. INTRODUCTION

Document forgery fraud presents one of the major problems across various industries, as important documents carrying financial records, legal contracts, or even identification papers can be put at risk. Moreover, techniques of forgery continue to change with time; hence, various technologies, from digital to handwritten ones, make it hard even for experts to distinguish what is real and what is tampered with. This further transcends the limitation of manual forgery detection techniques, complicated by the reality that paper records are nowadays increasingly being digitized for sharing purposes across businesses and industries. As such, several researchers have proposed machine learning and computer vision methods for the development of more accurate forgery detection systems, especially through CNN.[1].

This paper, on the other hand, explains a hybrid multi-task learning approach with expert labels for CNN-based segmentation in document forgery detection. Traditionally, forgery has been understood to mean the actual, physical tampering with documents. In digital forensics and document analysis, CNNs have found many helpful applications in various deep learning architectures of tasks related to images, including object detection, segmentation, and classification. A CNN consists of layers of connected nodes, just like the fashion in which the human brain would process any given visual information. It is important to note that the detection of document forgery can be done as an image segmentation problem, whereby the model learns to find suspicious document regions that could have been tampered with, leveraging the capabilities of CNNs. Another important concept in multi-task learning is when a model is jointly trained on related tasks, resulting in better generalization and improved performances. This means that, other than locating forged areas, the CNN identifies and segments textual elements, enhancement in the detection ability concerning forgeries in both handwritten and digital documents [2].

AI inventions have significantly advanced forgery detection of forged documents based on methods such as Capsule Networks (CapsNet) in order to preserve spatial hierarchies for subtle changes. Methods such as Error Level Analysis, approaches based on the RGB color channel, and GLCM texture descriptors proved quite effective, especially when combined with CNN-based offline signature verification. Difficulty lies in detecting smaller, more finely tampered regions and generalizing the model for different formats, resolutions, and layouts of the document. In particular, generally speaking, more research is apparently required for the improvement of distortion and complexity while maintaining cross-domain generalization in detection in document forgery.[3].

AI innovations like Capsule Networks (CapsNet) tend to significantly improve forged document detection by preserving spatial hierarchies to perceive the subtle changes between the original and the forged version. Other techniques, such as Error Level Analysis, approaches based on the color channels of RGB, and GLCM texture descriptors are promising but rather simple and mainly applicable in CNN-based offline signature verification [2]. This paper proposes a hybrid model based on CNN that addresses the problems of the challenges related to image and text forgery detection. The model is well-equipped with the ability to detect various manipulated regions in images, text, and other elements in the document making the model pretty adaptable to a wide range of documents, both digital and paper-based. Multi-task learning enables the model to run multiple tasks simultaneously, thus enhancing the model's ability to detect sophisticated forgeries. Unlike image and text-based models exclusively, this model relies both on images and text to detect the tampered visual parts-photos or stamps-while also identifying text anomalies-altered characters or forged signatures. As the forgery detection can be viewed as an image segmentation task, the model segments a document into meaningful segments for more accurate analysis, especially if differences between the original and the forged parts of a document are minimal. It also provides adequate accuracy for fine-grained manipulations at low resolutions or degraded images. Shared learning among tasks also helps develop computational efficiency and scalability, making the model suitable for high-volume applications in finance and government sectors, where it has massive documents to process. It also promotes cross-model learning abilities across different document formats, resolutions, and layouts. Although progress has been achieved in this area, much research is still needed to reduce distortion effects and cross-domain generalization while maintaining low complexity for document forgery detection.

The proposed CNN model architecture would thus ensure high accuracy for the proposed CNN model architecture, which is very lightweight and deployable for any real-time forgery detection applications, without imposing any significant computational burden. Most importantly, the emphasis on interpretability and transparency by the model addresses the common problem of deep learning models being black boxes. With the use of expert labels to produce bounding boxes around suspicious regions, this model provides explainable reasons for its decisions, which plays an important role in high-stakes applications, such as verification of legal documents and the detection of financial fraud. This architecture applies the technique of multi-task learning, identify text along with text rendering in problematic images. It detects forgeries in digital and handwritten documents by applying state-of-the-art AI methods to develop a robust detection technique for forensic applications. [4].

II. LITERATURE SURVEY

Author(s)	**Focus of the Paper**	**Key Points in Coverage**	**Technique(s) Used**
Nandini N, Keerthi Joshi K, Devprakash, Madhura C, Vandana M Ladwani [5]	Document Forgery Detection using AI and Machine Learning.	Introduction to capsule neural networks and error level analysis for forgery detection and, addressing challenges in detecting forgeries in various document formats.	Capsule Neural Network (CapsNet), Error Level Analysis (ELA), CNN, OpenCV for preprocessing
Chenfan Qu, Chongyu Liu, Yuliang Liu, Xinhong Chen, Dezhi Peng, Fengjun Guo, Lianwen Jin [6]	Detecting tampered text in document images using the Document Tampering Detector (DTD) framework.	Emphasizes improving detection robustness by addressing inconspicuous tampering in visually consistent areas. It introduces a new dataset, DocTamper, for evaluation.	Frequency Perception Head (FPH), Multi-view Iterative Decoder (MID), Curriculum Learning for Tampering Detection (CLTD), and the use of DCT coefficients
Ismail Taha Ahmed, Baraa Tareq Hammad, Norziana Jamil [7]	CMFD with 3 levels of Ward linkage based clustering.	Robustness: JPEG compression, translation, scale. Limitations: Not robust to rotation, blur, contrast change, small duplicated region.	Keypoint-Based Methods using Local Symmetry and Random Sampling Consensus (RANSAC)
Taha Barwahwala, Aprajit Mahajan, Shekhar Mittal, Ofir Reich [8]	Challenges of using machine learning to improve tax collection in developing countries.	Examines the effectiveness of a machine learning model in identifying fraudulent firms for tax purposes in India, highlighting that despite the model's accuracy, institutional barriers prevented significant improvements in tax collection.	Random Forest Classifier, PageRank, K-core Decomposition, Fused Transaction Network Representation, ATTENet
S. S. Gornale, G. Pattil, and R. Benne [9]	Use of RGB color channels and GLCM texture descriptors to identify forged handwritten and printed documents.	Method found to be effective for distinguishing between genuine and forged documents, achieving 95.8% and 93.11% accuracy rates for forged handwritten and printed documents respectively	RGB color components, GLCM texture descriptors, Support Vector Machine (SVM)

III. METHODOLOGY

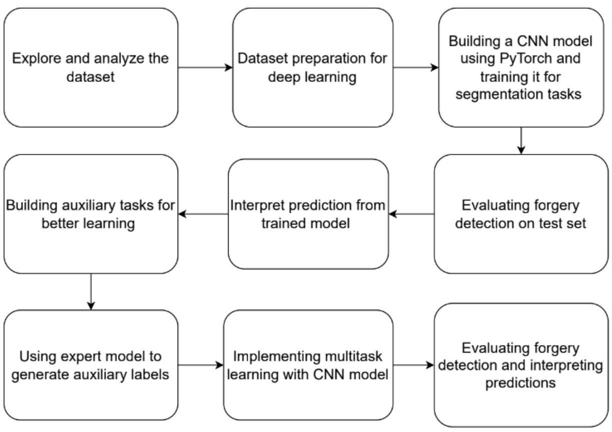

Fig. 1 Flow Diagram for the methodology adopted

1) Explore and analyze the dataset

In the model, the dataset used for training is divided into three main folders: 1) train, 2) val, and 3) test. These, in turn, have a list of image files. Corresponding to each of these, there exists a text file, namely, train.txt, which lists, row by row, forgery information for each of these images contained within that particular set. In addition, within each folder, there will be a combination of both genuine and forged receipt images.

2) Dataset preparation for deep learning

Dataset preparation will include the following processes: create a PyTorch Dataset object for the dataset, have receipt images organized, and also be prepared with the corresponding labels, for instance, bounding box annotations over forged regions. Then load it through a DataLoader to efficiently perform batching, shuffling, and parallel processing. Then create segmentation masks from the bounding box annotations. This will result in each pixel being classified as either "forged" or "genuine," essentially reducing the problem to pixel-level classification.

3) Building a CNN model using PyTorch and training it for segmentation tasks.

This project implements a PyTorch CNN for image segmentation, aimed at the detection of forged document regions. Concretely, it has been adopted in this context with the DeepLabV3 architecture, recently widely adopted in literature for pixel-wise classification challenges. This model will classify every pixel in an image as "forged" or "genuine." Below is the instantiation of the DeepLabV3 model in PyTorch with its backbone from MobileNetV3, giving it a very light feature extraction capability while remaining quite powerful. There will be two classes of output, namely one for the genuine and another for the forged. The training here will leverage the GPU if that is available for faster computation. The main challenge, class imbalance, is an issue because the number of forged regions is usually much fewer compared with the number of genuine ones in the dataset. A weighted cross-entropy loss function is applied, such that higher weights are assigned to the minority forged class to prevent the model from being biased toward genuine pixels. This allows the CNN to focus on the appropriate identification of forged regions and therefore improves performance and robustness for document forgery detection tasks.

4) Evaluating forgery detection on test set

Essentially, the evaluation of forgery detection on a test set involves generalizing the performance of the model on new data. This model was put in evaluation mode, model.eval(), during which it should not update any parameters. The model predicts on test set images of receipts and their respective forgery segmentation masks, comparing the same against the ground truth labels. The most common evaluation metric is the test set loss that is usually computed by the weighted cross-entropy loss function in order to take into consideration the class imbalance problem between forged and genuine pixels. The lower the test set loss, the better the performance, and this value will be compared to losses during training and validation in order to check overfitting. Ideally, test set loss would be as close as possible to the validation loss, signaling it picked up generalizable patterns. Besides the important inspections, understanding how well the model detects forged regions through the use of heatmaps; quantitative metrics are also utilized. These heatmaps overlay the model's forgery probability predictions onto the original images and give insight into the regions of an image where the model is focusing its effort in detection. There are some discrepancies in the prediction between the training and test sets, whereby further refinement of the model should be considered to help it in that regard for effective generalization. Multi-task learning could be pursued to this end.

5) Interpret prediction from a trained model.

Generally, predictions from a model trained for forgery detection need to have some sense of understanding from how it points at the forged region. Normally, these types of predictions are visualized as heat maps where the probability of forgery detection is overlaid on top of the original receipt images. The darker regions in the heat map will result in high confidence for detecting forged regions, and vice-versa. It would be nice if there were some comparison between those heatmaps and the ground truth forgery regions-annotated by bounding boxes-to assess how well the model is doing. Mismatches between the predicted and actual forged regions can indicate areas where the model struggles or makes errors and may account for the performance of the model, hence emphasizing other aspects which may need refinement. This in turn forms the required interpretation, thereby translating into useful insight to develop the thought process for model performance enhancement.

6) Building auxiliary task for better learning

Better learning by incorporating an auxiliary task involves having a secondary task complement the main task in model performance improvement. For some contexts, this could mean document forgery detection-where tampered regions in receipts are the primary interest and where text detection could serve as a highly useful auxiliary task. A possible reason for this could be that forgery usually affects textual elements like the price or amount in words/numbers and detects changes that provide additional context to identify forged areas. The authors incorporate an auxiliary text detection task using a pre-trained text detection model. This expert model produces bounding boxes on text regions in receipt images that are subsequently used to create additional labels for the dataset. With these labels instead of manually annotating text regions, which are automatically coming from the text detection model inserted into the dataset with the forgery annotations, this also helps ease the pains in data preparation and utilizes the already available advanced models for augmenting the dataset. A main forgery detection model gets enriched training information with this text detection label inclusion regarding the textual content. It would also provide broader context to the model to learn patterns related to both text and forgery, which usually improves the accuracy and robustness regarding the detection of tampered regions. Hence, this auxiliary text detection task actually serves the primary task of forgery detection much better.

7) Using the expert model to generate aux labels

To generate auxiliary labels for text detection and improve document forgery detection, an expert model is adopted. It is impossible to manually label the text regions in receipt images in case of a large dataset; hence, utilizing a pre-trained text detection model downloaded from the Hugging Face model hub is the alternative. An expert model trained on extensive text data processes receipt images and outputs heat maps that show the existence of text with different confidences. Those heatmaps are further processed to identify and delineate the text regions within the images. The identified regions are converted into bounding boxes serving as pseudo-ground truth labels for the text detection task. These auxiliary labels will be combined with forgery labels and used to train a multi-task CNN model that learns to segment images into background, forged regions, and text regions. This approach leverages existing capabilities in the expert model for the automation of labeling, integrates auxiliary tasks without hassle, and thereby improves overall performance in forgery detection.

8) Multi-Task learning with CNN model

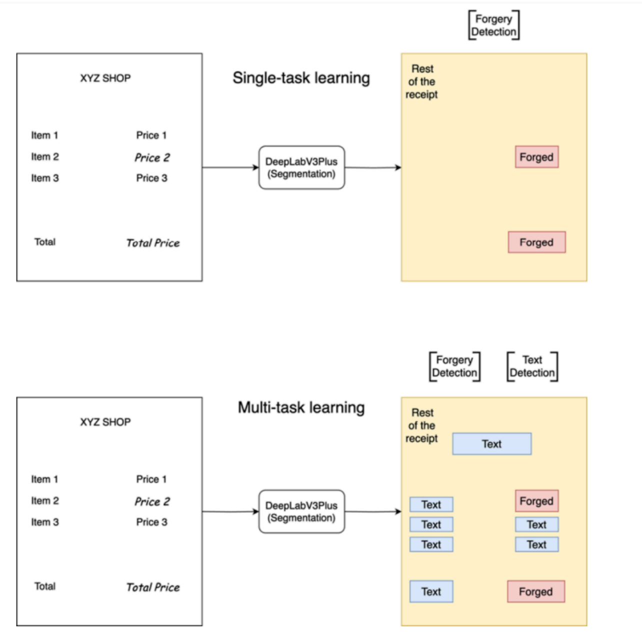

Fig. Single task learning (forgery detection) vs multi task learning (forgery detection + text detection)

It trains a single model on several related tasks simultaneously, leveraging synergy across tasks. Narrowly speaking, any adaptation of CNN models, such as DeepLabV3, would be meant for forgery and text detections regarding document forgery detection. This model targets the detection of forged regions within the images of receipts in order to specify exactly the area where tampering has taken place. An auxiliary task designed for text detection can be jointly trained by location and recognition of the texts from receipts. This is an important auxiliary task because it can provide some very valuable contextual information of the text through detection and thus further improve the performance of the model for forgery detection. The labels from both tasks will, therefore, be utilized for the training of the CNN model: forgery annotations include bounding boxes around the detected forged regions, while text detection labels are produced by an expert model. The expert model is pre-trained on text detection tasks and provides heat maps showing the presence of possible text regions in receipts. These heatmaps will then be converted to bounding boxes in order to create the labels for text detection. Combined with forgery annotations, it will learn to handle both problems jointly. In general, such shared learning between forgery detection and text detection improves overall performance because the model is allowed to take advantage of the interrelated nature of these tasks in such a way that knowledge learned in one task helps the other by improving insight and accuracy. This multitask learning, therefore, not only clears the way in single-model training but also enhances the model's effectiveness in individual models which may have been trained on diverse tasks.

9) Evaluating Forgery Detection and Interpreting Predictions

The only prerequisite to carry out the evaluation of this multi-task forgery detection model is a test set with receipt images which it has never seen. Calculate the loss on the test set. Use a weighted cross-entropy loss, identical to that of training. If this test set loss turns out to be much lower compared to the first training losses, then that is a sign of better generalization. If not, then high test loss may be indicative of overfitting. Further studies provide more detailed insights by generating heatmaps from the model predictions that visualize the confidence of the model in identifying forged regions. The higher the probability, for instance, the brighter the color on the heatmap, the higher the likelihood of forgery. These heatmaps, when compared to ground truth, give a better insight into the accuracy of the model and help identify problems such as overfitting or class imbalance. While the multitask model generally outperforms, its heatmaps are normally better aligned with actual forged areas; at the same time, further improvements might be necessary when fine-tuning model architecture and possibly integrating other tools such as OCR.

IV. RESULTS AND DISCUSSION

A) Digital forgery detection

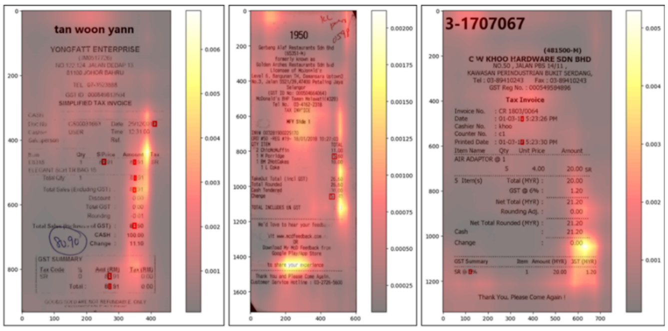

Fig. 3 The forged regions have been marked on the images that will be used for training using the image annotations. This trains the machine learning model to scan only those areas that are more suspicious. Bounding boxes have also been created for the same.

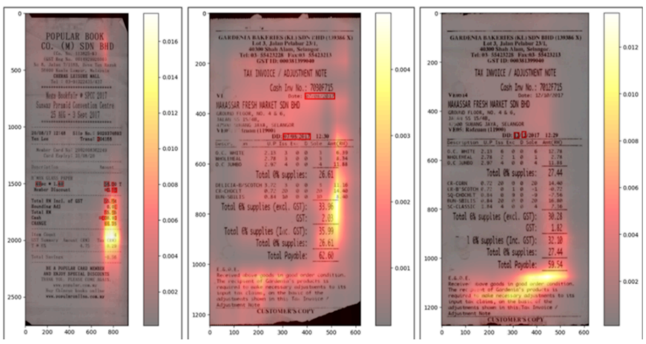

Fig. 4 The performance of machine learning model is tested by providing the test images

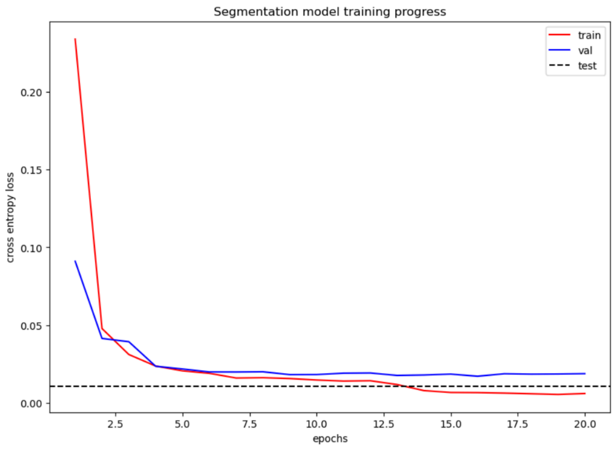

Fig. 5 Learning curves for the segmentation-based forgery detection model show how the training and validation losses progress along the training epochs. The dotted line shows the test set performance (loss) of the trained model.

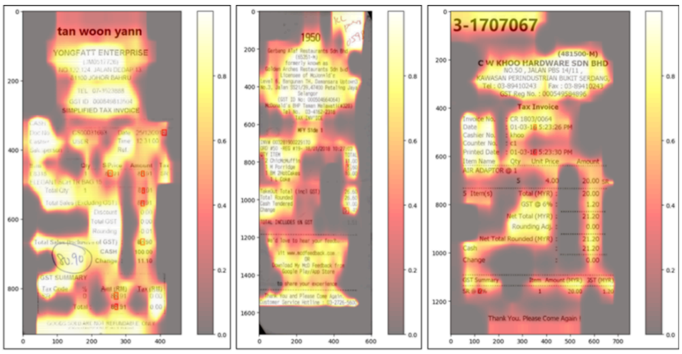

Fig. 6 Training images of multitasking machine learning model which denoted the probability heatmaps of where we want the model to look at during forgery detection

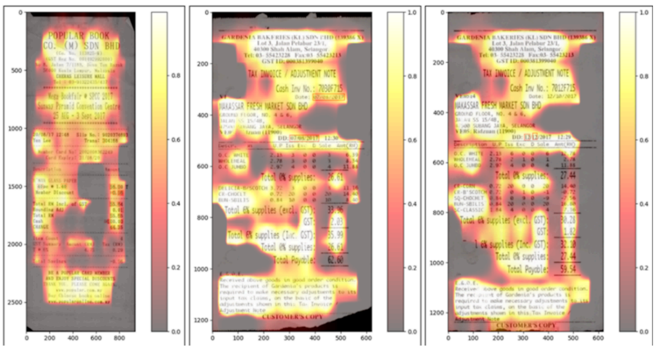

Fig. 7 The performance of machine learning model is tested by providing the test images

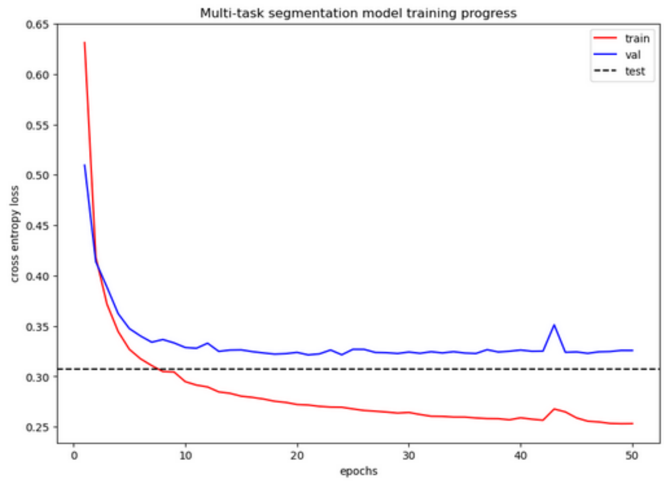

Fig. 8 Learning curves for the multi-task learning model show how the training and validation losses progress along training epochs. The dotted line shows the test set performance (loss) of the trained model.

Fig. 9 Image of receipt with bounding boxes drawn over text which is predicted by (expert) text detection model

B) HANDWRITTEN FORGERY DETECTION

Fig. 20 A total of 5 signatures have been taken from each individual to train the machine learning model

TO TEST THE MACHINE LEARNING MODEL, 4 SIGNATURES OF DIFFERENT INDIVIDUALS HAVE BEEN PROVIDED

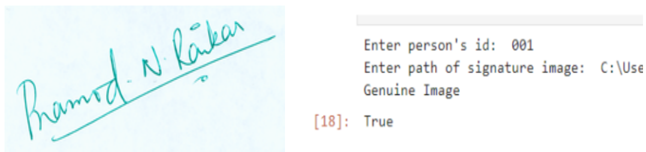

Fig. 11 Signature of A done by A

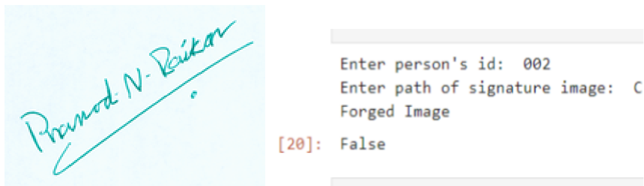

Fig. 12 Signature of A done by B

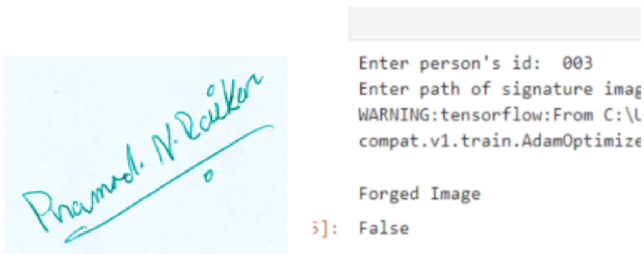

Fig. 13 Signature of A done by C

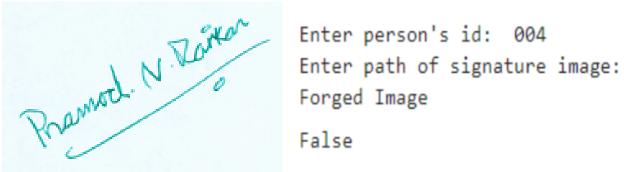

Fig. 14 Signature of A done by D

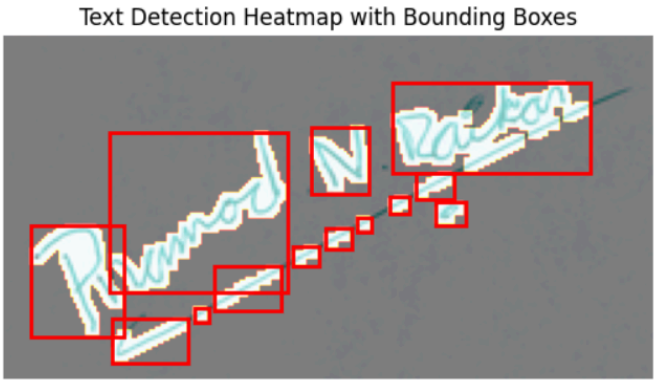

Fig. 15 Identification of suspicious region with heatmap and bounding boxes

V. COMPARISON WITH OTHER METHODOLOGY

TABLE I

Sr no.	**Comparison of Document Forgery Detection Techniques and their Performance**		
	Paper	**Methodology**	**Accuracy**
1	Mohamed Sirajudeen and R. Anitha [10]	Convolutional Neural Network (CNN)	85.09 %
2	Shruti Ranjan et al. [2]	Artificial Neural Networks (ANN), Linear SVM	96.4 %
3	Lin Zhao et al. [11]	Text Editing Network (ForgeNet)	Not specified
4	Zanardelli et al. [12]	Various Deep Learning Approaches	Not specified
5	Model used in this paper	Deep Learning, Hybrid and Multitasking based Document Forgery Detection	90 - 95 %

VI. CONCLUSION AND FUTURE SCOPE

The proposed hybrid model of CNNs and multi-task learning for document forgery detection has come up with a strong arm against the growing terror of document fraud in its digital and handwritten manifestations. Since the model treats forgery detection as an image segmentation problem, it hence identifies forged regions with high precision. Employing multi-task learning enhances the process of detection by simultaneously recognizing textual and visual elements in documents. The dual capability, when combined with expert labels, will not only increase the accuracy but also enhance interpretability-manually ensuring that stakeholders understand why certain areas have been flagged as tampered. The hybrid model, due to its scalability and lightweight architecture, is suitable for applications in real time from various sectors involving voluminous document processing, such as financial institutions and government agencies. This is very useful in the detection of forgeries, especially within low-resolution or degraded images, which is very common in document scanning and compression.

Further development of the architecture must be infused with transformer-based segmentation models in order to enhance forgery detection accuracy. The dataset needs to be augmented by fine-tuning hyperparameters for generalization across document types and formats. Moreover, Optical Character Recognition for consistency checks will add another layer of validation, especially in the case of numerical data such as receipts.

This can also be further researched with the incorporation of deep learning models, such as GANs, capable of simulating forgery techniques to produce synthetic forgeries. This will widen the datasets and expose the model to a wide variety of forgery types, thus making the model more robust. These could also be enhanced with the use of attention mechanisms in the models, which narrow down the focus to parts of a document that are most relevant, as minute and localized regions may not easily be detected. The detection, when explained through combination with XAI techniques, provides transparency in the process so that insights on why a certain area has been classified as forged can be derived more intuitively.

The goal of this paper has been the introduction of a hybrid method for document forgery detection with CNNs and multi-task learning, which detects image and text-based forgeries and, at the same time, increases the subtlety of the tampered document regions that can be detected with improved interpretability of expert labels. Moving forward, we will continue to enhance our model using advanced techniques, such as XAI, while ensuring our detection process is highly accurate and transparent, with reasons for the decisions made by the model provided back to the users.

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